Minor Project Synopsis

on

AQUA: AI-based Quality Understanding and Analytics

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Project Guide

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AQUA: AI-based Quality Understanding and Analytics

A Machine Learning and Full-Stack Framework for Water Quality Forecasting and Early Warning

1. Literature Review and Research Gap

Water quality monitoring is essential for environmental protection and sustainable management of surface-water resources. Conventional water-quality assessment generally relies on periodic collection of samples followed by laboratory analysis. Although such methods provide reliable measurements, they are time-consuming and do not always provide continuous insight into changes in water quality.

Machine Learning (ML) and Deep Learning (DL) techniques have increasingly been applied to water-quality prediction and classification. Recent studies have explored models including Random Forest, Support Vector Machines, Gradient Boosting, LSTM and other deep-learning approaches for predicting parameters such as dissolved oxygen, pH and other water-quality indicators. Reviews published in recent years show that ML can capture complex relationships between water-quality parameters and environmental variables and can support prediction, classification and monitoring [2].

Recent research has also explored the use of ML for anomaly detection and early identification of unusual water-quality conditions. Explainable ML is another emerging area, as understanding the factors influencing a prediction is important when ML models are used for environmental decision-making [3].

However, existing research is often focused on individual prediction models or individual water-quality parameters. Challenges also remain in integrating spatial information, temporal information and environmental variables into a unified prediction framework. Data availability and differences in sampling frequency further complicate the development of reliable water-quality prediction systems. Recent literature identifies the need for improved integration of heterogeneous environmental data, robust evaluation and interpretable ML approaches [4].

For India, the National Water Data Portal provides CPCB surface-water datasets containing station information, geographic coordinates, sampling dates and multiple physical, chemical and biological water-quality parameters. The portal provides separate datasets for different temporal characteristics, including annual chemical measurements and monthly physical and biological measurements [5].

Research Gap

Based on the reviewed literature, there is scope for developing an integrated framework that combines water-quality observations with spatial, temporal and relevant environmental information, while also evaluating model behaviour across different monitoring locations.

AQUA aims to address this gap by investigating:

- integration of water-quality, spatial and environmental data;

- comparison of suitable ML models for water-quality prediction;
- evaluation of model performance across monitoring stations rather than relying only on conventional random train-test splits;
- detection of unusual water-quality behaviour and potential risk conditions;
- use of explainability techniques to understand important prediction factors; and
- deployment of the resulting ML system through a full-stack monitoring platform.

The exact prediction targets and temporal modelling approach will be finalized after detailed analysis of the available CPCB and environmental datasets.

2. Objective and Problem Statement

Problem Statement

Existing water-quality monitoring largely depends on periodic sampling, making it difficult to continuously assess changing water conditions and identify potential deterioration at an early stage. At the same time, available water-quality datasets contain information at different spatial and temporal resolutions, making their effective integration and use for prediction challenging.

The proposed project aims to develop AQUA, an integrated Machine Learning and Full-Stack framework that uses available water-quality, spatial and environmental data to predict water-quality conditions, identify unusual patterns and provide early warnings for potentially deteriorating conditions.

Objectives

The major objectives of AQUA are:

- To collect and preprocess publicly available surface-water quality data for selected monitoring stations in the Yamuna basin.
- To integrate water-quality observations with spatial, temporal and relevant meteorological/hydrological variables.
- To develop and compare suitable ML models for predicting selected water-quality parameters.
- To investigate whether models trained using data from existing monitoring stations can generalize effectively to other monitoring locations.
- To identify anomalous or potentially deteriorating water-quality conditions using prediction errors and/or relevant water-quality thresholds.
- To apply explainability techniques to identify important factors influencing model predictions.
- To develop a full-stack platform for visualizing station-wise water-quality information, predictions, trends, anomalies and alerts.

3. Methodology

The proposed methodology will consist of the following stages:

3.1 Data Acquisition

Surface-water quality data will primarily be obtained from CPCB datasets available through the National Water Data Portal. Relevant meteorological variables will be obtained from suitable publicly available historical weather sources. Hydrological variables such as river discharge may also be incorporated where suitable data coverage and compatibility are established.

Initial feasibility testing has already demonstrated that CPCB station coordinates and sampling dates can be successfully matched with historical weather data from Open-Meteo/ERA5-based data.

3.2 Data Preprocessing

The collected datasets will be cleaned and standardized by:

- handling missing and inconsistent values;
- converting dates into a common format;
- removing duplicate observations;
- standardizing station identifiers;
- validating geographical coordinates;
- analysing parameter-wise data completeness; and
- selecting suitable target parameters based on data availability.

The different datasets will then be aligned spatially and temporally.

3.3 Feature Engineering

Potential features will include:

- historical water-quality measurements;
- latitude and longitude;
- month and season;
- temperature;
- precipitation/rainfall;
- relative humidity;
- relevant hydrological variables such as river discharge, where available;
- lagged and rolling environmental variables.

The final feature set will be determined after exploratory analysis of the available data.

3.4 Machine Learning Modelling

The existing Random Forest model will be used as an initial baseline.

Additional suitable models such as XGBoost/LightGBM and, if supported by the temporal density of the final dataset, appropriate time-series/deep-learning models may be evaluated.

Models will be compared using appropriate regression/classification metrics depending on the selected prediction task.

3.5 Spatial Generalization

Instead of relying only on a conventional random train-test split, the project will investigate station-wise generalization.

Data from selected monitoring stations will be used for training while another station will be kept unseen during training and used for evaluation. This will help determine whether the model can generalize to locations that were not included in its training data.

3.6 Anomaly Detection and Early Warning

The system will identify unusual behaviour using model prediction errors and/or relevant water-quality thresholds.

Where applicable, CPCB Designated Best Use criteria will be used to interpret water-quality conditions. CPCB defines Classes A–E using parameters including pH, dissolved oxygen, BOD and other criteria depending on the designated use [6].

The system will generate alerts when predicted or observed conditions indicate potentially deteriorating water quality or significant deviations from expected behaviour.

3.7 Explainability

Explainability techniques such as SHAP (SHapley Additive exPlanations) will be explored for suitable models to identify the variables contributing most strongly to predictions.

3.8 Full-Stack Implementation

A full-stack web platform will be developed to integrate the ML system into a usable monitoring interface.

The platform will include:

- Frontend: interactive dashboard, station map, trends and alerts;
- Backend: APIs for retrieving station information, predictions and alerts;
- Database: storage of station information, water-quality observations, environmental data and model outputs;
- ML integration: connection between the trained models and the backend;
- Visualization: station-wise predictions, historical trends, risk status and explainability information.

4. Plan

Weeks	Planned Work
1–2	Literature review, problem formulation, dataset collection and exploratory analysis
3–4	Data cleaning, preprocessing and integration of water-quality and environmental data

Weeks	Planned Work
5–6	Feature engineering and development of baseline ML models
7–8	Model comparison, evaluation and spatial generalization experiments
9	Anomaly detection, early-warning mechanism and model explainability
10–11	Backend, database and ML API development
12	Frontend integration, testing and development of the initial AQUA prototype

Note: *The exact prediction targets, environmental variables and final ML models will be finalized after detailed analysis of dataset completeness and temporal/spatial compatibility.*

5. References

[1] National Water Informatics Centre, Surface Water Quality (Manual – Chemical Parameters), CPCB, National Water Data Portal.

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[3] "A review of machine learning and internet-of-things on the water quality assessment: Methods, applications and future trends," Results in Engineering, 2025.

[4] "A Bibliometric-Systematic Literature Review (B-SLR) of Machine Learning-Based Water Quality Prediction: Trends, Gaps, and Future Directions," Water, 2025.

[5] National Water Informatics Centre, Surface Water Quality (Manual – Physical Parameters), CPCB, National Water Data Portal.

[6] Central Pollution Control Board, Designated Best Use Water Quality Criteria.

[7] Helaly, M. A., et al., "Advancements in water quality prediction: a practical review of machine learning and deep learning approaches," Cluster Computing, 2025.

[8] "Deep learning for water quality," Nature Water, 2024.